

TorqCapture — Rulebook (standalone event)

Grade: absorbs "Capture the Zone" (was Best) · **Role:** TorqWars' team-vs-team territory game — the whole day is a mode-rotation playlist · **Basically:** Call of Duty objective modes with RC cars — capture the region, not the flag

*Built 2026-09-06 from [[TorqCapture — Design Study v1]] (the locked concept). TorqCapture reframes the old flag-run "Capture the Zone" into FPS-style territory control: you don't carry an object, you **hold ground**. Cars can't shoot, so the whole game turns on one mechanic — **occupy a glowing zone, out-number the enemy inside it, shove theirs out, hold the majority till it flips**. Electronic scoring is locked ([[TorqCapture — Zone-Capture Tech]]) is the prototype gate). Reuses the TorqFall base kit (rented fleet, turf, foam bump-rings, barriers, show layer) + TorqSumo's push mechanic + a small zone kit. Same standalone-event frame as TorqFall/TorqSiege — its own day, its own room, nothing shared.*

One floor, a **playlist of 5 modes** swapped in minutes like a COD session. Two teams of coloured cars fight over lit rings; the tech counts who's inside, calls the capture, flips the ring's colour and ticks the score to the big screen. Team colours, a live map of who's winning, and a last-second contested flip are the content engine.

1. Objective

Two teams fight for control of one or more marked **zones** on the floor. You **capture** a zone by getting more of your cars inside it than the enemy has, and **holding** that majority for a short timer — the zone then flips to your colour and **ticks points** for as long as you keep it. The enemy drives in to take it back; you **body-check them out of the ring** to defend it. Most points (or most zones held) at time wins. No shooting, no carrying — just **who owns the ground, right now**.

2. The core mechanic (count → hold → flip → contest)

This is the whole game. Every zone is always in exactly one of two states — **NEUTRAL (uncaptured)** or **CAPTURED (owned)** — and the rules below define precisely what "pauses" in each, so there's no ambiguity between "nobody's scoring yet" and "the owner keeps scoring."

- **Occupy = hold.** A zone is a marked ring, ~1.5–2 m across (§3). A car with your team's tag physically inside the ring counts as one of your cars "on the ground" there.
- **NEUTRAL zone — no owner, no points, only the hold-timer is live.** An uncaptured ring scores nobody yet. **Live majority** = more of your cars inside the ring than the enemy has, right now. Hold that live majority continuously for the **~5 s hold-timer** (the default; tuned at the bench test — [[TorqCapture — Zone-Capture Tech]]) and the ring **CAPTURES**: it flips to your colour and starts ticking points for you. **Equal count, or an empty ring, only pauses the hold-timer** — it does not reset it, and nothing is ticking either way because nobody owns the ring yet. Re-establish the live majority and the paused timer resumes from where it left off.

- CAPTURED zone — ownership + colour LATCH.** Once a ring has flipped, it **keeps ticking points for its owner every second, even if it goes empty or unattended**, until the enemy both (a) gets a **live majority** inside it, **and** (b) holds that majority for a **fresh ~5 s flip-timer**. **An equal count in a captured zone only pauses the enemy's flip-timer — it never stops the owner's points, which keep ticking throughout the contest.** This is exactly what makes "own 2 of 3 and roam to contest the third" (Domination, §4a) work: the two zones you already hold keep banking points for you whether or not you have a car standing in them.
- Contest = shove them out.** A rival drives in to break your majority — in a neutral ring, to stop your hold-timer; in a captured ring, to start (or hold back) their flip-timer. You **bump them out of the ring** — the "weapon" is the bumper, this is **TorqSumo's push mechanic aimed at an objective**. Push the count back in your favour and the enemy's timer pauses again; your own points (if you're the owner) were never at risk.
- Lose it = their flip-timer completes.** If the enemy holds a live majority in a zone you own for the full fresh flip-timer, the ring flips to their colour — ownership and points transfer, and it's now their latched zone to defend. Hold them back out (shove, or pile more cars in) before their timer completes and you keep the zone.

Zone-seconds (used in §7) = seconds a zone is OWNED, not seconds a car is physically present in it. A captured zone banks zone-seconds for its owner every second it's held, whether or not the owner has a car standing in it — the mechanic behind Domination's "hold 2, roam to contest the third."

The feel: a glowing ring, your two cars inside it holding the line, an enemy car nosing in — one of yours peels off to shove it back out while the other keeps the count. The ring pulses your colour, the screen ticks up. That is TorqCapture.

3. Format

- Teams: 2 teams, 2–3 cars each** (default **2 × 3 = 6 cars on the floor**; drop to **2 × 2 = 4 cars** for single-zone modes when the ring would jam). This reuses TorqFall's **6–8-car heat model** — the fleet, charging rotation and pit are identical (see [[TorqFall — What The Event Needs]] §1); a TorqCapture "match" is just those 6–8 cars split into two coloured sides instead of a free-for-all pack.
- Team colour = your tag.** Every car carries a strap-mounted team-coloured tag/beacon (§4). Red vs Blue is the default; the tag is what the zone reader counts and what the overhead cam sees flip.
- Match length: 2–6 min depending on mode** (per-mode timing in §4; The Hill's quick filler round is the sub-3-min exception at 2–3 min) — long enough for the lead to swing 2–3 times, short enough to keep the queue moving. A points cap ends a match early if one team runs away with it.
- Roster / rotation:** teams are formed at registration (walk-ins drafted into Red/Blue, or friend-groups kept together). Losing a match never ends a driver's day — non-elim qualifying, same as TorqFall (see the day arc below). Wrecked-out or waiting drivers cycle to free-play laps between matches; nobody stands around.

The day structure — the playlist arc (a mode rotation, the headline feature; shuffle it every event so no two days repeat):

Block	Modes	Format	Elimination
Open qualifying (bulk of the day)	Domination heats, with a Hotzone heat mixed in for energy	2×3, round-robin; teams earn points by match result + zones held; leaderboard on the screen	None — non-elim. Every team gets N matches (default 3); a paid ticket never ends in one round.

Block	Modes	Format	Elimination
Bracket (afternoon)	Domination → Siege Point (once the crowd knows the game)	Top teams seeded; match winners advance	Winner advances; loser drops to free-play + consolation laps.
Grand Final (the climax)	The Hill	One central zone, both finalist teams, hold longest	The showpiece — one ring, one crowd, a single contested timer everyone watches.

- **Between blocks:** free-play open laps + a quick **Courier** exhibition (the legacy flag-run) as a palate-cleanser while crew re-lay the zones. Nobody waits in silence.
- **Mode swap = a 2–5 min crew layout change** (move/retape zones, load the mode preset on the controller) — allowed because we own the floor all day; within a match nothing moves.

4. The five modes (the playlist)

Each mode is the core mechanic (§2) on a different zone layout and win condition. Spec table first, then per-mode detail.

Mode	COD parent	Zones	Win condition	Length	Teams	When to run
Domination (default)	Domination	3 zones A/B/C across the floor	Most zone-seconds at time, or first to the points cap	5–6 min	2 × 3	The base mode — bulk of qualifying + the bracket
Hotzone	Hardpoint	1 zone that relocates every 60–90 s	Most zone-seconds across all locations	~5 min	2 × 2–3	A high-energy heat mixed into qualifying; the scramble content
The Hill	King of the Hill	1 central zone	Team holding it longest at time	2–3 min (quick) / up to 5 (final)	2 × 2–3	The fast filler round AND the grand final
Siege Point (advanced)	Headquarters	1 zone; on capture it locks for N s	Most locked-captures banked	~6 min	2 × 3	The bracket, once the crowd reads the game
Courier (legacy)	payload / flag-run	1 neutral core + each team's home zone	Most deliveries of the core to your zone	~5 min	2 × 2–3	A between-blocks exhibition — the old game, kept as flavour

4a. Domination (default)

- **Layout:** 3 zones — **A / B / C** spread across the floor (A and C near each team's end, B central — the contested middle). The floor splits into a front line.
- **Rule:** hold a zone = it ticks points/sec for you. **Own 2 of 3 = you're winning the tug-of-war**; the enemy must break your middle or flank a corner. All three can be flipping at once — the reader tracks each independently.

- **Win:** highest zone-seconds at the 5–6 min horn, or first to a points cap (e.g. 200). A team that grabs all 3 briefly gets a bonus tick.
- **Team play:** 2×3 lets a team run "2 defend a held zone, 1 roams to contest" — the readable classic. Best base mode; the one a walk-in learns on.
- **When:** the default. Bulk of qualifying and most of the bracket.

4b. Hotzone

- **Layout:** **ONE lit zone.** Every **60–90 s it relocates** to a new marked spot on the floor (the controller lights the next ring; pre-taped spots, so no crew on the floor).
- **Rule:** same count/hold/flip, but the whole pack must **scramble across the floor** every relocation — abandon the ring you were holding and race for the new one. Holding through a relocation is impossible; the fight resets each time.
- **Win:** most zone-seconds banked across all locations at ~5 min.
- **Content:** the **relocation scramble is the money frame** — 6 cars breaking for a new ring at once, on the overhead cam. Commentary writes itself ("Hotzone's moved to the far corner — Blue's caught out!").
- **When:** one heat dropped into qualifying to spike the energy; a crowd favourite.

4c. The Hill

- **Layout:** **ONE central zone**, dead middle of the floor.
- **Rule:** simplest in the set — capture it and hold; the clock counts your held-seconds. No flanking, no second objective, just a single ring both teams pile into and shove over.
- **Win:** the team holding it **longest** at time. In the quick version, first to X held-seconds.
- **When:** two jobs. (1) The **quick filler** round (2–3 min) when the queue is short. (2) The **grand final** (up to 5 min) — one ring, both finalist teams, the whole crowd watching a single contested timer. The purest, most legible finish in the game.

4d. Siege Point (advanced)

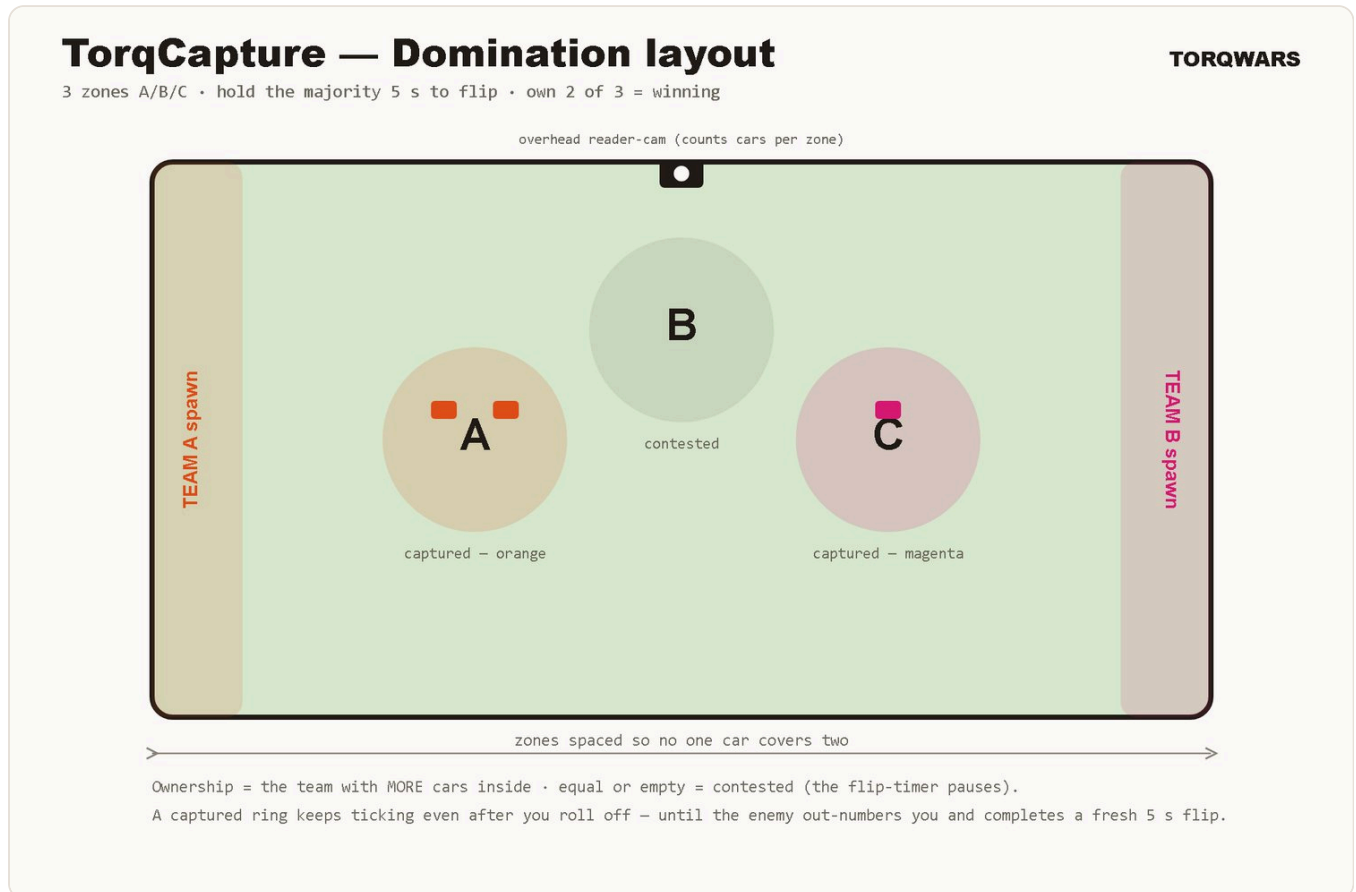
- **Layout:** ONE zone at a time. On capture, the zone **locks** — it can't be retaken for **N s** (e.g. 30 s) — then it **despawns and a new one spawns** elsewhere.
- **Rule:** adds a **defend phase**. Attackers race to capture; the instant they flip it, it locks and their points are banked safe for N s while the defenders can only reposition for the next spawn. Then it moves and the roles reset — an **assault / defend rhythm**, not a constant tug-of-war.
- **Win:** most locked-captures banked at ~6 min.
- **Why advanced:** the lock rule needs the crowd to already understand capture — save it for the bracket, never a walk-in's first match.
- **When:** later bracket rounds.

4e. Courier (legacy — the absorbed flag-run)

- **Layout:** one **neutral core** (a lit ball/object) at centre + each team's **home zone** at their end.
- **Rule:** the old **Capture the Zone** game, kept as ONE mode. A car nudges/carries the neutral core to its home zone to score a delivery; the core respawns at centre; contest by shoving the carrier. (Carry method = push the core, or a simple cradle — decided at build; no bolting to rented cars.)
- **Win:** most deliveries at ~5 min.

- **Why it survives:** TorqCapture **absorbs and supersedes** Capture the Zone (locked decision); the fetch variant lives on here as flavour, not the whole game. Runs as a between-blocks exhibition so the legacy game still gets a spotlight.

5. The arena



- **Floor:** the shared **~3000 sq ft** event floor (100–300 people), same envelope as TorqFall/TorqSiege. Play area turfed; spectators behind a soft barrier line. Not all 3000 is turfed — the play zone + runoff is (see [[TorqCapture — What The Event Needs]] for the turf buy).
- **Zones (the rings):** each zone is a **~1.5–2 m ring** marked on the turf. Three build options, decided at the bench test:
 1. **Flat taped ring** — coloured gaffer circle, the reader/antenna at its edge or an overhead cam defining it in software. Cheapest, nothing for a car to catch on.
 2. **Raised foam bump-ring** — a low pool-noodle/EVA lip around the circle (like the car bump-rings), so a car nudged to the edge **rebounds back in** and holds position better. Helps the "hold" feel; tested for whether it traps cars.
 3. **Lit ring** — a WS2812B LED strip under a diffuser at the ring edge that **changes colour on capture** (the content hero — see §8). Combine with 1 or 2.
- **Zone spacing (on 3000 sq ft):** in **Domination**, A/B/C sit ~4–6 m apart so a car takes a few seconds to re-deploy between them — far enough that you can't hold two at once, close enough to scramble. Single-zone modes (Hotzone/Hill) use pre-taped spots across the same floor. Keep every ring ≥ 2.5 m off the crowd barrier.

- **Team colour tags:** each car carries a strap-mounted **team-coloured tag/beacon** (velcro/zip-tie, nothing bolted — rented fleet). This is both what the **reader counts** and what the **overhead cam sees**. Red vs Blue default; a third colour (Green) exists for a future 3-team variant.
- **The reader + scoreboard:** each zone is read by **an overhead-cam region (primary), or a UHF RFID antenna (fallback)** — the tech is DECIDED in [[TorqCapture — Zone-Capture Tech]] (CV primary, UHF fallback; BLE/RSSI was evaluated and ruled out as a primary because it can't draw a reliable 1.5–2 m boundary). The reader feeds a controller (ESP32/laptop) that runs **count → majority → hold-timer → capture → LED colour + live score to the big screen**. A **marshal is backup for disputes only** — the tech is the scorer.
- **Soft side barriers** the full perimeter (noodle-on-sandbag rail, TorqFall stock) so a shoved car bounces back in, not into the crowd. Foam bump-rings on every car (§7).

6. Match flow (marshal + tech script)

1. **Load the mode.** Tech loads the mode preset on the controller (zone map, hold-timer, relocation interval, points cap); the correct rings light up.
2. **Stage.** 4–6 cars staged at their team's start spot; drivers on the drivers' stand, Red one side, Blue the other. Marshal confirms every car's tag reads on the board (a "tag check" — no dead beacons on the floor).
3. **Launch.** Marshal countdown "3-2-1-GO"; both teams break for the zone(s).
4. **Run.** Cars pile into rings, shove for the majority, hold. **The tech scores it live** — counts per team per zone, hold-timers, flips, ticking points, all on the big screen. Marshal watches for fouls (§7) and works disputes; a second marshal (or the MC) calls the swings for the crowd.
5. **Relocation (Hotzone) / lock (Siege Point):** the controller drives these automatically — the ring relocates or locks on its own timer; the marshal just confirms the floor is clear of a stuck car at the new spot.
6. **Contested calls.** Equal count / empty ring = contested by the tech automatically — the **active** timer pauses (the hold-timer on a neutral ring, or the enemy's flip-timer on a zone you already own; an owned zone's points never pause, §2). The marshal only intervenes if a car is flipped, jammed, or a tag drops out — calls "HOLD", clears it, resumes.
7. **Finish.** At the horn (or the points cap) the tech calls the winner to the screen; the winning zone(s) pulse the winner's colour. Marshal confirms, MC announces.
8. **Reset.** Marshals collect cars to the start, swap in charged cars, tech re-arms the mode — **next match in ~30–60 s** (same-mode) or a **2–5 min layout change** between blocks (mode swap).

7. Scoring, safety & fouls

Scoring:

- **Within a match:** zone-seconds (points ticked per second held) and/or deliveries (Courier) / locked-captures (Siege Point), all tallied by the tech. The screen is the source of truth.
- **Qualifying (non-elim):** teams earn leaderboard points by match result + zones held; best N matches count. A lost match is a low score, never an exit.
- **Bracket / final:** match winner advances; The Hill final decides the day's champion team (trophy + podium — the closing reel).

Safety (same envelope as TorqFall):

- **Speed-limited cars only** (~30 km/h cap, per [[Car Fleet, Scale & Protection]] §3–4). A 4–6-car shove-fest at capped speed is safe; at full brushless speed it's a demolition derby into the crowd line.
- **Foam bump-rings** zip-tied on every car (come off clean — rented fleet); soft perimeter barrier full length; spectators behind it, ≥ 2.5 m off any ring.
- **Shoving is the only "combat"** and it is body-to-body at low speed — everyone can do it, nobody feels bad losing.

Fouls (marshal's call, no review — throughput depends on fast calls):

- **Ramming a car already out of the zone:** the shove is to clear a rival **from the ring**; once a car is fully out, **leave it** — chasing it down outside the ring is a foul (skip that car's next contest / a warning). Keeps the fight *on the objective*, not a wrecking derby.
- **Spoofing the count:** deliberately parking a dead car, or defeating the reader to fake presence/absence — **occluding or blocking a car's overhead marker** (the CV-primary exploit: markers ride on top, visible to the cam, so hiding one drops it or a rival from the count) **or shielding/masking a UHF tag** (the fallback exploit) — = that car pulled for the match.
- **Crossing into a locked zone (Siege Point):** driving into a zone that's locked to the enemy = pushed back out by the marshal, no penalty (it just doesn't count) unless repeated.
- **Leaving the play area / cutting across the crowd side of a barrier:** car pulled, no points that match.
- **Ramming after the horn / spoiling a reset:** driver skipped a turn.
- **Marshal is final.** Disputes go to the marshal, who checks the tech; the tech's count stands unless it's visibly wrong (a dropped tag), which is exactly what the marshal is there to catch.

8. Content cues (the day's money shots)

- **Overhead cam locked on the floor** — the zones **flip colour on camera** as control changes: a live, legible map of who's winning. The single best shot in the game and the reason CV scoring doubles as the content feed ([[TorqCapture — Zone-Capture Tech]]).
- **The Hotzone relocation scramble** — the whole pack breaking for a new ring at once. Cut every one for reels.
- **The last-second contested flip** — a hold-timer counting down while an enemy car noses into the ring; a teammate shoves it out at 1 s to save the capture. The signature clip.
- **Team colours pop on camera** — Red vs Blue is built-in narrative + built-in cheering. The only true team-vs-team game in the set — the most *social* thing we run.
- **Big screen = the scoreboard** — ticking points, flipping zones, the lead swinging. The MC calls it live ("Blue's lost B! they're scrambling for the middle!").
- **Onboard FPV** on 1–2 cars for the "storming the point" POV; **low ring-level cam** for the shove-out.

9. Variants (pocket rules — programme by shuffling modes, don't invent on the day)

- **3-team (Red/Blue/Green):** Domination with a third colour on the tags — a three-way for a bigger crowd. Needs the reader to distinguish 3 tag colours (CV does this free; RFID/BLE need 3 tag IDs).
- **Overtime:** a tied Hill or Domination goes to a sudden-death single ring — first capture wins.
- **Payload Courier:** the core must be delivered along a lane (a moving objective) instead of straight home — the closest thing to a COD payload.

- **Double-cap:** two Hotzones lit at once, worth more together — splits a 3-car team.
- **Handicap:** a stronger team plays 2 cars vs the other's 3 — the count mechanic makes handicapping trivial and fair.

10. Reset / turnaround (for run-of-show)

- **Between matches (same mode):** collect cars + swap charged + re-arm the mode = **~30–60 s**. The throughput engine.
 - **Between blocks (mode swap):** move/retape zones + load the new mode preset = a **2–5 min crew layout change** — covered by free-play laps, the Courier exhibition + the MC.
 - **Within a match nothing moves** (Hotzone relocation and Siege lock are software, not crew) — the floor is built once on build morning.
 - **Tag/battery check every match** (the marshal's stage-check) + a reader sanity-check every ~30 min so the scorer never drifts.
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Pairs with: [[TorqCapture — Zone-Capture Tech]] (the reader/tag/controller build + the prototype gate — do this before anything) · [[TorqCapture — What The Event Needs]] (fleet, zone kit, show layer + the full costed needs list — its quantities derive from §3–§5 here) · [[TorqCapture — Design Study v1]] (why territory-hold, why moded) · [[Event Ops Standards (all games)]] · [[Car Fleet, Scale & Protection]] (the rented fleet + speed cap + foam rings this reuses). **Absorbs** [[Capture the Zone]] — the flag-run survives as the **Courier** mode (§4e); Capture the Zone is superseded in the index/games.csv. TorqFall, TorqSiege and TorqBall are their **own separate events** — nothing here shares a day, a floor or a schedule with them.

Verified + changelog (2026-09-06): first build from the locked Design Study v1. Speced all 5 modes (Domination/Hotzone/The Hill/Siege Point/Courier) with layout, win condition, timing, team sizes + when-to-run; nailed the core count→hold→flip→contest mechanic; set team sizes to 2×2–3 reusing TorqFall's 6–8-car heat model; speced the arena (1.5–2 m rings, three ring-build options, A/B/C spacing on 3000 sq ft, colour tags, reader→scoreboard); wrote the marshal+tech match script, scoring/safety/fouls (incl. the "no ramming a car already out of the zone" rule), content cues, variants, reset times. Hold-timer (~5 s), ring size, relocation interval and raised-ring question all flagged as bench-test tunables (see Zone-Capture Tech). Absorbs Capture the Zone as Courier per the locked decision. **Correction (2026-09-06, examiner pass 2):** aligned the Rulebook with the DECIDED tech in [[TorqCapture — Zone-Capture Tech]] — §5 now reads "an overhead-cam region (primary), or a UHF RFID antenna (fallback)" and notes BLE/RSSI was ruled out as a primary (it can't hold a reliable 1.5–2 m boundary), instead of listing UHF/BLE/CV as three co-equal live options; §7's spoofing foul is generalised to cover both builds — occluding/blocking a car's overhead marker (the CV-primary exploit, since markers are visible on top) OR shielding/masking a UHF tag (the fallback exploit). **Correction (2026-09-06, examiner pass 3):** §2 was self-contradictory — it said an empty/equal-count ring "pauses" the timer in the same breath as describing a captured zone that should keep ticking points while unattended, so a reader couldn't tell whether "pauses" meant the capture clock or the owner's score. Rewritten as two explicit states: a **NEUTRAL** zone (no owner, no points) where only the ~5 s hold-timer runs and pauses on equal count/empty ring; a **CAPTURED** zone where ownership + colour LATCH and points keep ticking every second regardless of occupancy, until the enemy gets a live majority AND completes a fresh ~5 s flip-timer (equal count there only pauses THEIR flip-timer, never the owner's score). Added an explicit "zone-seconds = seconds owned, not seconds present" definition (feeds §7), and reworded §6.6's contested-call line and the "pauses" bullet to match. This is what makes Domination's

"hold 2, roam to contest the third" (§4a) actually work as written. Also §3: match length corrected from "3–6 min" to "2–6 min depending on mode" — The Hill's quick round is 2–3 min, below the old 3 min floor; noted as the sub-3-min exception.